Full Stack Assignment – 21

Assignment ID: FSD-21

Title: Internal Knowledge Base Management System

Duration: 3 Hours

1️⃣ Objective

Design and develop an **Internal Knowledge Base Management System** that allows contributors to submit articles and administrators to control publication.

The system must enforce approval before publishing and ensure **data persistence using a database**.

2️⃣ User Roles

CONTRIBUTOR

- Register and log in to the system
- Create knowledge articles
- View own articles and their status

ADMIN

- Log in to the system
- Review submitted articles
- Approve or reject articles for publishing

3️⃣ Functional Requirements

Authentication & Authorization

- Implement user registration and login
- Use **JWT-based authentication**
- Protect all backend APIs using JWT
- Enforce **role-based access control** on APIs and frontend views

Article Management

- CONTRIBUTOR can create articles with title and content
- Articles must be stored in the database
- ADMIN can approve or reject articles
- Only approved articles should be visible as published

4️⃣ Business Rule (Critical)

⚠️ **Articles must be approved before they can be published or viewed publicly.**

Unapproved articles must not be visible to other users.

5️⃣ Technical Constraints (Mandatory)

- All article data **must persist in a database**
- Hardcoded or in-memory storage is **not allowed**
- Backend APIs must reflect actual database state
- JWT authentication and role validation are mandatory
- Use environment variables for sensitive configuration

6️⃣ Frontend Requirements

- Login and registration pages
- Role-based dashboards for CONTRIBUTOR and ADMIN
- Article creation form for CONTRIBUTOR
- Article review and approval view for ADMIN
- Published article listing view

UI design is not evaluated; workflow correctness matters.

7️⃣ Submission Requirements (Strict)

GitHub Repository

- Backend source code
- Frontend source code
- Proper folder structure
- Clear and meaningful commit history

Live Deployment

- Live frontend deployment link
- Live backend deployment link
- Backend must be connected to the database

README.md (Mandatory)

The README file must include:

- Project overview
- Tech stack used
- User roles and permissions
- API endpoints

- Database schema or collections
- Steps to run the project locally
- Live frontend and backend links

8️⃣ Evaluation Note

Evaluation focuses on **approval workflow enforcement, role handling, and end-to-end execution**.